AI-Based Intelligent Bus Arrival and Scheduling System

Technical Research Report

Indha report AI, IoT, GPS, machine learning matrum dynamic scheduling-ai use panni intelligent public bus transportation system-ai design seyyum muraiyai explain seigiradhu.

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Abstract

Traditional bus systems fixed timetable-ai depend pannuvadhaal traffic congestion, rain, accidents, passenger crowd matrum road delays nala actual arrival time schedule-oda match aagama pogiradhu. Indha report, real-time GPS, IoT sensors, traffic data, weather data, passenger demand matrum historical trip records-ai combine panni bus arrival time predict seyyum AI-based intelligent system-ai propose seigiradhu. Machine learning model ETA, delay matrum passenger demand-ai predict seyyum; optimisation engine bus holding, extra-bus deployment, vehicle allocation matrum dynamic schedule adjustment-ai recommend seyyum. Indha approach passenger waiting time, bus bunching matrum overcrowding-ai reduce panni public transport reliability-ai improve panna uthavum.

Keywords

Artificial Intelligence, Machine Learning, GPS, IoT, ETA Prediction, Bus Scheduling, Bus Bunching, GTFS Realtime, Reinforcement Learning, Smart Transportation.

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1. Introduction

Namma daily transportation-la bus service oru mukkiyamana public transport mode. Aana traditional bus system mostly fixed timetable basis-la operate aagum. Traffic congestion, signal delay, road construction, rain, accidents, passenger boarding time matrum vehicle breakdown maadhiri factors nala bus actual arrival time schedule-oda match aagama irukkum.

Idhunaala passengers-ku bus eppo varum-nu correct-a theriyama unnecessary waiting time increase aagum. Oru bus late-a varumbodhu adutha bus-oda serndhu “bus bunching” nadakkalaam. Transport operator-ku vehicle allocation, timetable revision matrum driver planning kashtama irukkum.

Indha problem-ai solve panna AI-Based Intelligent Bus Arrival and Scheduling System propose pannappadugiradhu. Indha system GPS, IoT sensors, traffic data, historical trip data, weather data matrum machine learning algorithms-ai use panni bus ETA predict pannum; real-time situation-ku according schedule-ai dynamically optimise pannum.

Research Problem

Fixed schedule-based system actual traffic and passenger conditions-ai properly consider pannaadhu. Enave, real-time bus location, traffic, passenger demand matrum historical pattern-ai analyse panni accurate arrival prediction and dynamic scheduling provide seyyum system thevai.

Objectives

• Bus-oda current location-ai GPS moolama track seyyaradhu.

• Ovvoru bus stop-kum Estimated Time of Arrival calculate seyyaradhu.

• Traffic, weather, passenger crowd matrum historical data-ai consider seyyaradhu.

• Delay-ai early-a identify seyyaradhu.

• Bus bunching-ai reduce seyyaradhu.

• Passenger waiting time-ai minimise seyyaradhu.

• Demand-ku suitable-a bus frequency and vehicle allocation-ai optimise seyyaradhu.

• Passenger app, web dashboard matrum bus-stop display-la live information show seyyaradhu.

2. Research Idea and Concepts

2.1 System Concept

Ovvoru bus-um GPS tracker, mobile GPS application illa IoT tracking unit moolama location data send pannum. Indha data cloud server-ku pogum. Server historical data, traffic, weather matrum passenger data-oda combine panni AI model-kku input kudukkum. AI model current location, next stop, ETA, expected delay, passenger demand matrum scheduling recommendation-ai output-a generate pannum.

2.2 Important Terms

Estimated Time of Arrival (ETA): Oru bus specific bus stop-ku arrive aagum expected time. New GPS data varumbodhu ETA continuous-a update aaganum.

Headway: Rendu consecutive buses-kulla irukkura time gap. Headway romba small-a irundha bunching; romba large-a irundha passenger waiting time increase aagum.

Bus bunching: Oru bus delay aagi adhiga passengers-ai pickup pannumbodhu dwell time increase aagum. Adutha bus first bus-ai catch up pannumbodhu rendu buses same time-la varum phenomenon bus bunching.

Holding: Bus-ai control point-la konjam neram hold panni buses-kulla even gap maintain seyyum control method.

3. Existing System and Limitations

• Fixed timetable and manual dispatch.

• Driver experience-based decisions.

• Static passenger-demand estimation.

• Periodic route surveys mattum.

• Real-time bus position visibility kuraivu.

• Sudden traffic or rain conditions handle panna mudiyama irukkum.

• Bus bunching and long service gaps.

• Future delay predict panna mudiyadhu.

• High-demand route-kku immediate-a extra bus allocate panna kashtam.

Existing systems present status-ai display panninaalum proactive prediction and optimisation-ai generally provide pannaadhu. Indha gap-ai proposed AI system address seyyum.

4. Proposed Solution

Proposed system oru AI-powered real-time decision-support platform. Idhu passenger application mattum illa; transport operator-kku fleet-management, prediction and scheduling system-um aagum.

4.1 Input Data

• GPS: latitude, longitude, speed, direction, timestamp, vehicle ID and trip ID.

• Route data: route number, stop sequence, stop coordinates and normal travel time.

• Historical data: previous arrival/departure time, route-wise delay, peak-hour pattern and dwell time.

• Traffic data: congestion, road speed, accidents, road closures and signal delays.

• Passenger data: boarding, alighting, occupancy, queue length and ticketing data.

• Weather data: rainfall, temperature, visibility and storm/flood alerts.

• Calendar data: weekday, weekend, holiday, festival and special events.

4.2 Outputs

• Bus current location and next stop.

• Stop-wise ETA.

• Delay classification.

• Passenger demand level.

• Bus bunching probability.

• Holding or dispatch recommendation.

• Dynamic schedule update.

• Passenger notification.

5. System Architecture

System architecture multi-layer approach-ai follow pannum. Data acquisition-lendhu passenger application varai each layer specific responsibility-ai handle pannum.

Architecture Flow: GPS/IoT Devices → Communication Layer → Real-Time Data Ingestion → Data Cleaning and Map Matching → Historical and Real-Time Database → AI Prediction Engine → Scheduling Optimisation Engine → Passenger App, Admin Dashboard and Bus Stop Display.

5.1 Data Acquisition Layer

Bus-la installed GPS tracker regular interval-la location send pannum. Passenger counter, electronic ticketing device, IR sensor illa camera occupancy and boarding data collect panna mudiyum.

5.2 Communication Layer

4G/5G, Wi-Fi, MQTT, HTTP REST API illa GSM module use panni data server-kku transmit pannalaam. Network unavailable aana device local-a data cache panni connection restore aana piragu upload panna vendum.

5.3 Data Ingestion Layer

Incoming GPS and sensor streams receive, validate, duplicate remove, timestamp check and correct processing service-kku forward seyyappadum.

5.4 Data Cleaning and Map Matching

Invalid coordinates, duplicate records, abnormal speed and missing values handle seyyappadum. GPS point nearest road segment and predefined bus route-oda match pannappadum.

5.5 Storage Layer

PostgreSQL/PostGIS spatial data-kku, TimescaleDB time-series GPS-kku, Redis current ETA cache-kku, cloud object storage raw and historical data-kku use panna mudiyum.

5.6 AI and Optimisation Layers

AI prediction layer ETA, delay, demand and bunching probability calculate pannum. Optimisation layer dispatch, holding, short-turning, route reallocation and schedule adjustment recommendations-ai generate pannum.

6. AI Models and Algorithms

6.1 ETA Prediction

Historical average simple baseline-a use pannalaam. Linear regression distance, congestion, weather and time factors-ai model pannum. Random Forest or XGBoost non-linear tabular relationships-ai handle pannum. LSTM/GRU previous GPS sequence and delay pattern-ai learn panna useful. Road network and connected stops-ai graph-a represent panna Graph Neural Network use panna mudiyum.

Hybrid strategy: Normal conditions-la XGBoost; temporal pattern-ku LSTM; GPS missing aana historical average; sudden incident aana traffic-alert correction.

ETA calculation: ETA_stop = CurrentTime + sum(TravelTime_i + DwellTime_i). Dwell time passenger boarding and alighting count-ku depend aagum.

6.2 Passenger Demand Prediction

Stop ID, route ID, time, day, holiday, weather, nearby school/office timing and historical boarding data use panni demand low, medium or high-nu predict pannalaam. XGBoost, LSTM or temporal forecasting model use panna mudiyum.

6.3 Delay Prediction

Current progress-ai expected progress-oda compare panni delay estimate pannappadum. Delay classes: on-time, minor, moderate and major delay.

6.4 Reinforcement Learning

Reinforcement learning-la state as bus location, delay, headway, passenger queue, occupancy, traffic and available buses represent pannappadum. Actions as continue, hold, dispatch, short-turn, reassign or schedule change. Reward function passenger waiting time, schedule deviation, bunching, operating cost and overcrowding-ai minimise panna design pannappadum.

7. System Workflow

1. Bus registration: vehicle ID, route ID, GPS device and trip details assign pannappadum.

2. Data collection: GPS, speed, timestamp and passenger data collect pannappadum.

3. Real-time transmission: Network moolama central server-kku send aagum.

4. Validation: Coordinates, timestamp, speed and route assignment check pannappadum.

5. Map matching: Bus current road segment and next stop identify pannappadum.

6. Feature generation: Distance, speed, traffic, weather, delay and demand features create pannappadum.

7. ETA prediction: Upcoming stops-kku AI arrival time calculate pannum.

8. Continuous correction: New GPS update varumbodhu ETA recalculate aagum.

9. Bunching detection: Consecutive bus headway target-oda compare pannappadum.

10. Scheduling decision: Hold, dispatch, reallocate or short-turn recommendation generate aagum.

11. User notification: Passenger app, display and dashboard update aagum.

12. Performance logging: Predicted and actual arrival compare panni model improve pannappadum.

8. Scheduling Optimisation

Scheduling engine multiple objectives-ai balance panna vendum: passenger waiting time, schedule deviation, bus bunching, operating cost and overcrowding.

Objective: Minimise Z = w1W + w2D + w3B + w4C + w5O, where W is waiting time, D schedule deviation, B bunching penalty, C operating cost and O overcrowding penalty.

Possible Actions

• Holding: Too-early bus-ai control stop-la short time hold seyyaradhu.

• Extra bus deployment: High-demand prediction irundha reserve bus dispatch seyyaradhu.

• Short-turning: Full route delay adhigama irundha intermediate stop-la bus turn back seyyaradhu.

• Vehicle reallocation: Low-demand route-lendhu high-demand route-kku spare bus move seyyaradhu.

• Dynamic timetable: Expected delay based on passenger-facing schedule update seyyaradhu.

Constraints

• Available bus count.

• Driver duty hours.

• Bus capacity and maintenance status.

• Fuel or battery level.

• Depot location.

• Minimum and maximum headway.

• Route coverage and road restrictions.

9. Passenger Application and Operator Dashboard

9.1 Passenger Application

• Live bus tracking.

• Stop-wise ETA.

• Route search and source-destination selection.

• Occupancy indication.

• Delay and cancellation alerts.

• Alternative route suggestion.

• Tamil and English language support.

• Voice announcement and accessibility support.

9.2 Administrator Dashboard

• All buses live map.

• Route-wise bus count and delay list.

• Bus bunching alerts.

• Passenger demand heatmap.

• Predicted versus actual ETA graph.

• Vehicle and driver status.

• Schedule recommendation.

• Route punctuality report.

10. Database and Technology Stack

10.1 Main Database Tables

Table	Important fields
Vehicle	vehicle_id, registration_number, capacity, gps_device_id, status
Route	route_id, route_name, origin, destination, distance, duration
Bus Stop	stop_id, stop_name, latitude, longitude, sequence
GPS Logs	vehicle_id, latitude, longitude, speed, timestamp, route_id
Prediction	vehicle_id, stop_id, predicted_arrival, actual_arrival, error, model_version

10.2 Recommended Technology

• Hardware: GPS tracker, ESP32/Raspberry Pi gateway, passenger counter, 4G/5G module and LED display.

• Backend: Python, FastAPI/Flask, PostgreSQL/PostGIS, Redis, MQTT/Kafka and Docker.

• Machine learning: Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow or PyTorch.

• Frontend: React/Angular dashboard and Flutter/React Native mobile app.

• Maps and monitoring: Leaflet/Mapbox, Prometheus and Grafana.

11. Model Evaluation

Time-series data-ku random split avoid panna vendum. Old dates training, recent dates validation and latest dates testing-ku use panna vendum.

• MAE: Average ETA prediction error in minutes.

• RMSE: Large prediction errors-ku greater penalty.

• MAPE: Relative percentage error.

• On-time performance: Tolerance-kulla arrive aagura trips percentage.

• Passenger waiting time: Implementation-ku munnaadi/pinnaadi average comparison.

• Bus bunching rate: Target headway-kku below-a cluster aagura events.

• Occupancy and overcrowding: Overcrowded trips reduction.

• System latency: GPS update-lendhu ETA generation varaikkum time.

12. Security, Challenges and Implementation

12.1 Security and Privacy

- GPS data encrypted transmission.
- API authentication and role-based access.
- Passenger personal data anonymisation.
- Secure cloud storage, audit logs and backups.
- Camera data unnecessary-a long-term store panna koodadhu.
- Clear consent and privacy notice for mobile location data.

12.2 Challenges

- GPS error in urban buildings and tunnels.
- Network failure and missing data.
- Model drift after road or route changes.
- Incorrect vehicle IDs and poor data quality.
- AI recommendation explainability.
- Initial hardware, cloud and maintenance cost.
- Passenger counting sensor accuracy.

12.3 Implementation Plan

1. Requirement analysis and route digitisation.
2. One or two high-demand routes-la pilot deployment.
3. GPS, dashboard and historical dataset creation.
4. Baseline and advanced ETA model training.
5. Bunching detection and holding module development.
6. Offline replay, peak-hour, rain and network-failure testing.
7. Cloud deployment, user training, monitoring and periodic retraining.

13. Expected Advantages and Future Scope

13.1 Advantages

- Passengers-ku accurate ETA and less waiting time.
- Transport operators-ku better fleet utilisation and punctuality.
- Reduced bus bunching and overcrowding.
- Data-based timetable revision.
- Reduced unnecessary fuel consumption.
- Improved public transport trust and accessibility.

13.2 Future Enhancements

- Electric bus battery-aware scheduling.
- Traffic signal-la bus priority.
- Metro and train integration.
- Digital twin simulation.
- Predictive vehicle maintenance.
- Voice-based Tamil assistant.
- Emergency rerouting.

• Carbon emission estimation.

• Demand-responsive bus services.

14. Conclusion

AI-Based Intelligent Bus Arrival and Scheduling System public transport reliability-ai improve panna practical and scalable solution. GPS, IoT, historical data, traffic, weather and passenger demand-ai combine panni machine learning model accurate ETA generate panna mudiyum.

System-oda main benefit ETA display mattum illa. Dynamic scheduling, bus bunching control, passenger demand-based fleet allocation and proactive delay management moolama waiting time and overcrowding reduce panna mudiyum. Fixed timetable-ai replace pannama, AI-ai real-time correction and optimisation layer-a use pannuvadhu practical approach.

Pilot route-la system start panni, data quality improve seyyum podhu XGBoost, LSTM, graph model and reinforcement learning introduce panna vendum. Success-ai MAE, on-time performance, waiting time, bunching rate, occupancy, operating cost and passenger satisfaction moolama evaluate panna vendum.

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